

Calculus II

Trigonometric integrals

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Outline

- 1 Trigonometric Integrals
 - Integrating rational trigonometric integrals
 - Ad hoc methods for trigonometric integrals

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Integrals of the form $\int R(\cos \theta, \sin \theta) d\theta$, R

Let R be an arbitrary rational function in two variables (quotient of polynomials in two variables).

Question

Can we integrate $\int R(\cos \theta, \sin \theta) d\theta$?

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- Yes. We will learn how in what follows.
- The algorithm for integration is roughly:
 - Apply the substitution $\theta = 2 \arctan t$ to transform to integral of rational function.
 - Solve as previously studied.

The rationalizing substitution $\theta = 2 \arctan t$

Recall the expression of $\sin(2z)$, $\cos(2z)$ via $\tan z$:

$$\sin(2z) = ?$$

$$\cos(2z)$$

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Let R - rational function in two variables. $\int R(\cos \theta, \sin \theta) d\theta$ can be integrated via the substitution $\theta = 2 \arctan t$. How does this transform $\sin \theta, \cos \theta$?

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$d\theta$

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Theorem

The substitution given above transforms $\int R(\cos \theta, \sin \theta) d\theta$ to an integral of a rational function of t .

Example

$$\int \frac{d\theta}{2 \sin \theta - \cos \theta + 5}$$

Example

Let $\theta = 2 \arctan t$, $\cos \theta = \frac{1-t^2}{1+t^2}$, $\sin \theta = \frac{2t}{1+t^2}$

$$\int \frac{d\theta}{2 \sin \theta - \cos \theta + 5} = \int \frac{2dt}{(1+t^2) \left(2 \frac{2t}{t^2+1} - \frac{(1-t^2)}{1+t^2} + 5 \right)}$$

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$$\begin{aligned} \int \frac{d\theta}{2 \sin \theta - \cos \theta + 5} &= \int \frac{2dt}{(\textcolor{red}{1} + t^2) \left(2 \frac{2t}{t^2+1} - \frac{(1-t^2)}{1+t^2} + \textcolor{red}{5} \right)} \\ &= \int \frac{2dt}{6t^2 + 4t + \textcolor{red}{4}} \end{aligned}$$

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$$\begin{aligned} \int \frac{d\theta}{2 \sin \theta - \cos \theta + 5} &= \int \frac{2dt}{(1+t^2) \left(2 \frac{2t}{t^2+1} - \frac{(1-t^2)}{1+t^2} + 5 \right)} \\ &= \int \frac{2dt}{6t^2 + 4t + 4} \\ &= \int \frac{dt}{3t^2 + 2t + 2} \end{aligned}$$

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 \text{(complete square)} \quad &= \int \frac{dt}{3 \left(t^2 + 2t \frac{1}{3} + \frac{1}{9} - \frac{1}{9} + \frac{2}{3} \right)}
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 &= \frac{\sqrt{5}}{5} \arctan \left(\frac{3}{\sqrt{5}} \left(\tan \left(\frac{\theta}{2} \right) + \frac{1}{3} \right) \right) + C
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The integral $\int \sec \theta d\theta$ appears often in practice. A quicker solution will be shown later, but first we show the standard method.

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Set $\theta = 2 \arctan t$, $\cos \theta = \frac{1 - \tan^2(\frac{\theta}{2})}{1 + \tan^2(\frac{\theta}{2})} = \frac{1 - t^2}{1 + t^2}$.

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$$\begin{aligned} \tan \theta + \sec \theta &= \frac{\sin \theta + 1}{\cos \theta} = \frac{2 \sin(\frac{\theta}{2}) \cos(\frac{\theta}{2}) + \sin^2(\frac{\theta}{2}) + \cos^2(\frac{\theta}{2})}{\cos^2(\frac{\theta}{2}) - \sin^2(\frac{\theta}{2})} \\ &= \frac{(\sin(\frac{\theta}{2}) + \cos(\frac{\theta}{2}))^2}{(\cos(\frac{\theta}{2}) - \sin(\frac{\theta}{2})) (\cos(\frac{\theta}{2}) + \sin(\frac{\theta}{2}))} \end{aligned}$$

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- Examples to which our ad hoc techniques apply arise from integrals needed outside of the subject of Calculus II, so these techniques are important.
- The trigonometric integral we saw, $\int \frac{d\theta}{2 \sin \theta - \cos \theta + 5}$, will not work with any of following ad-hoc techniques, so the general method is important as well.

Example

$$\int \sin^3 x dx$$

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$$\begin{aligned}\int \sin^3 x dx &= \int \sin^2 x \sin x dx \\ &= \int \sin^2 x d(?) \end{aligned}$$

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When n – odd:

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Set $\sin x = u$

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 &= \int u^m (1 - u^2)^{\frac{n-1}{2}} du
 \end{aligned}$$

When n – odd:

$$\cos x dx$$

$$= d(\sin x)$$

Express $\cos x$
via $\sin x$

$$\text{Set } \sin x = u$$

$$\begin{aligned}
 \int \sin^m x \cos^n x dx &= \int \sin^{m-1} x \cos^n x d(-\cos x) \\
 &= - \int (1 - \cos^2 x)^{\frac{m-1}{2}} \cos^n x d(\cos x) \\
 &= - \int (1 - u^2)^{\frac{m-1}{2}} u^n du
 \end{aligned}$$

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If both m, n - even,

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 \end{aligned}$$

When m – odd:

$$\sin x dx$$

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Express $\sin x$
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$$\text{Set } \cos x = u$$

If both m, n – even, use $\left| \begin{array}{l} \sin^2 x = \frac{1 - \cos(2x)}{2} \\ \cos^2 x = \frac{\cos(2x) + 1}{2} \end{array} \right.$ and substitute $s = 2x$ to lower trig powers. Repeat above considerations.

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 \int \sin^m x \cos^n x dx &= \int \sin^m x \cos^{n-1} x d(\sin x) \\
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Express $\cos x$
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 \int \sin^m x \cos^n x dx &= \int \sin^{m-1} x \cos^n x d(-\cos x) \\
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Example

$$\int_0^{\frac{\pi}{2}} \sin^2 x dx$$

Example

Example

$$\int_0^{\frac{\pi}{2}} \sin^2 x \, dx = \int_0^{\frac{\pi}{2}} \left(\frac{1 - \cos(2x)}{2} \right) dx$$

express $\sin^2 x$
via $\cos(2x)$

Example

Example

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^2 x dx &= \int_0^{\frac{\pi}{2}} \left(\frac{1 - \cos(2x)}{2} \right) dx \\ &= \left[\frac{x}{2} - \frac{\sin(2x)}{4} \right]_0^{\frac{\pi}{2}}\end{aligned}$$

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$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^2 x dx &= \int_0^{\frac{\pi}{2}} \left(\frac{1 - \cos(2x)}{2} \right) dx \\ &= \left[\frac{x}{2} - \frac{\sin(2x)}{4} \right]_0^{\frac{\pi}{2}}\end{aligned}$$

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$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^2 x \, dx &= \int_0^{\frac{\pi}{2}} \left(\frac{1 - \cos(2x)}{2} \right) dx && \left| \begin{array}{l} \text{express } \sin^2 x \\ \text{via } \cos(2x) \end{array} \right. \\ &= \left[\frac{x}{2} - \frac{\sin(2x)}{4} \right]_0^{\frac{\pi}{2}} \\ &= \left(\frac{\pi}{4} - \frac{\sin \pi}{4} \right) - \left(0 - \frac{\sin 0}{4} \right) .\end{aligned}$$

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Example

$$\int_{t=0}^{t=1} \sqrt{1-t^2} \, dt$$

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Example

Set $t = \cos x$, $x \in [0, \frac{\pi}{2}]$.

$$\int_{t=0}^{t=1} \sqrt{1-t^2} dt$$

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Example

Set $t = \cos x$, $x \in [0, \frac{\pi}{2}]$. Then
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$$dt = d(\cos x) = -\sin x \, dx.$$

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 \int_{t=0}^{t=1} \sqrt{1-t^2} \, dt &= - \int_{x=\frac{\pi}{2}}^{x=0} \sqrt{1 - \cos^2 x} \sin x \, dx \\
 &= \int_{x=0}^{x=\frac{\pi}{2}} \sqrt{\sin^2 x} \sin x \, dx
 \end{aligned}$$

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 \int_0^{\frac{\pi}{2}} \sin^2 x dx &= \int_0^{\frac{\pi}{2}} \left(\frac{1 - \cos(2x)}{2} \right) dx && \left| \begin{array}{l} \text{express } \sin^2 x \\ \text{via } \cos(2x) \end{array} \right. \\
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Example

$$\begin{aligned}
 \int_0^{\frac{\pi}{2}} \sin^2 x \, dx &= \int_0^{\frac{\pi}{2}} \left(\frac{1 - \cos(2x)}{2} \right) dx && \left| \begin{array}{l} \text{express } \sin^2 x \\ \text{via } \cos(2x) \end{array} \right. \\
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 \end{aligned}$$

Example

Set $t = \cos x$, $x \in [0, \frac{\pi}{2}] \Rightarrow \sin x \geq 0$. Then
 $dt = d(\cos x) = -\sin x \, dx$.

$$\begin{aligned}
 \int_{t=0}^{t=1} \sqrt{1-t^2} \, dt &= - \int_{x=\frac{\pi}{2}}^{x=0} \sqrt{1-\cos^2 x} \sin x \, dx \\
 &= \int_{x=\frac{\pi}{2}}^{x=0} \sqrt{\sin^2 x} \sin x \, dx \\
 &= \int_0^{\frac{\pi}{2}} \sin^2 x \, dx
 \end{aligned}$$

Example

$$\begin{aligned}
 \int_0^{\frac{\pi}{2}} \sin^2 x \, dx &= \int_0^{\frac{\pi}{2}} \left(\frac{1 - \cos(2x)}{2} \right) dx && \left| \begin{array}{l} \text{express } \sin^2 x \\ \text{via } \cos(2x) \end{array} \right. \\
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 \end{aligned}$$

Example

Set $t = \cos x$, $x \in [0, \frac{\pi}{2}] \Rightarrow \sin x \geq 0$. Then
 $dt = d(\cos x) = -\sin x \, dx$.

$$\begin{aligned}
 \int_{t=0}^{t=1} \sqrt{1-t^2} \, dt &= - \int_{x=\frac{\pi}{2}}^{x=0} \sqrt{1-\cos^2 x} \sin x \, dx \\
 &= \int_{x=0}^{x=\frac{\pi}{2}} \sqrt{\sin^2 x} \sin x \, dx \\
 &= \int_0^{\frac{\pi}{2}} \sin^2 x \, dx = \frac{\pi}{4}.
 \end{aligned}$$

Example

$$\int \tan^8 x \sec^4 x dx$$

Example

$$\int \tan^8 x \sec^4 x dx = \int \tan^8 x \sec^2 x \sec^2 x dx$$

Example

$$\begin{aligned}\int \tan^8 x \sec^4 x dx &= \int \tan^8 x \sec^2 x \sec^2 x dx \\ &= \int \tan^8 x \sec^2 x d(?)\end{aligned}$$

Example

$$\begin{aligned}\int \tan^8 x \sec^4 x dx &= \int \tan^8 x \sec^2 x \sec^2 x dx \\ &= \int \tan^8 x \sec^2 x d(\tan x)\end{aligned}$$

Example

$$\int \tan^8 x \sec^4 x dx = \int \tan^8 x \sec^2 x \sec^2 x dx$$

$$= \int \tan^8 x \sec^2 x d(\tan x)$$

$$= \int \tan^8 x \left(? \right) d(\tan x)$$

Can we rewrite
 $\sec^2 x$ via $\tan x$?

Example

$$\begin{aligned}\int \tan^8 x \sec^4 x dx &= \int \tan^8 x \sec^2 x \sec^2 x dx \\ &= \int \tan^8 x \sec^2 x d(\tan x) \\ &= \int \tan^8 x (1 + \tan^2 x) d(\tan x)\end{aligned}$$

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Can we rewrite $\sec^2 x$ via $\tan x$?
Set $u = \tan x$

Example

$$\begin{aligned}\int \tan^8 x \sec^4 x dx &= \int \tan^8 x \sec^2 x \sec^2 x dx \\ &= \int \tan^8 x \sec^2 x d(\tan x) \\ &= \int \tan^8 x (1 + \tan^2 x) d(\tan x) \\ &= \int u^8 (1 + u^2) du \\ &= \int (u^8 + u^{10}) du\end{aligned}$$

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Example

$$\begin{aligned}\int \tan^8 x \sec^4 x dx &= \int \tan^8 x \sec^2 x \sec^2 x dx \\&= \int \tan^8 x \sec^2 x d(\tan x) \\&= \int \tan^8 x (1 + \tan^2 x) d(\tan x) \\&= \int u^8 (1 + u^2) du \\&= \int (u^8 + u^{10}) du \\&= ?\end{aligned}$$

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Set $u = \tan x$

Example

$$\begin{aligned}\int \tan^8 x \sec^4 x dx &= \int \tan^8 x \sec^2 x \sec^2 x dx \\&= \int \tan^8 x \sec^2 x d(\tan x) && \left. \begin{array}{l} \text{Can we rewrite} \\ \sec^2 x \text{ via } \tan x? \end{array} \right| \\&= \int \tan^8 x (1 + \tan^2 x) d(\tan x) && \left| \text{Set } u = \tan x \right. \\&= \int u^8 (1 + u^2) du \\&= \int (u^8 + u^{10}) du \\&= \frac{u^9}{9} + \frac{u^{11}}{11} + C\end{aligned}$$

Example

$$\begin{aligned}\int \tan^8 x \sec^4 x dx &= \int \tan^8 x \sec^2 x \sec^2 x dx \\&= \int \tan^8 x \sec^2 x d(\tan x) \\&= \int \tan^8 x (1 + \tan^2 x) d(\tan x) \\&= \int u^8 (1 + u^2) du \\&= \int (u^8 + u^{10}) du \\&= \frac{u^9}{9} + \frac{u^{11}}{11} + C\end{aligned}$$

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$$\begin{aligned}\int \tan^8 x \sec^4 x dx &= \int \tan^8 x \sec^2 x \sec^2 x dx \\&= \int \tan^8 x \sec^2 x d(\tan x) \\&= \int \tan^8 x (1 + \tan^2 x) d(\tan x) \\&= \int u^8 (1 + u^2) du \\&= \int (u^8 + u^{10}) du \\&= \frac{u^9}{9} + \frac{u^{11}}{11} + C\end{aligned}$$

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Example

$$\begin{aligned}
 \int \tan^8 x \sec^4 x dx &= \int \tan^8 x \sec^2 x \sec^2 x dx \\
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 &= \int \tan^8 x (1 + \tan^2 x) d(\tan x) && \text{Set } u = \tan x \\
 &= \int u^8 (1 + u^2) du \\
 &= \int (u^8 + u^{10}) du \\
 &= \frac{u^9}{9} + \frac{u^{11}}{11} + C \\
 &= \frac{\tan^9 x}{9} + \frac{\tan^{11} x}{11} + C .
 \end{aligned}$$

Example

$$\int \tan^5 x \sec^9 x dx$$

Example

$$\int \tan^5 x \sec^9 x dx = \int \tan^4 x \sec^8 x \tan x \sec x dx$$

Example

$$\begin{aligned}\int \tan^5 x \sec^9 x dx &= \int \tan^4 x \sec^8 x \tan x \sec x dx \\ &= \int \tan^4 x \sec^8 x d(\quad)\end{aligned}$$

Example

$$\begin{aligned}\int \tan^5 x \sec^9 x dx &= \int \tan^4 x \sec^8 x \tan x \sec x dx \\ &= \int \tan^4 x \sec^8 x d(\sec x)\end{aligned}$$

Example

$$\begin{aligned}\int \tan^5 x \sec^9 x dx &= \int \tan^4 x \sec^8 x \tan x \sec x dx \\ &= \int \tan^4 x \sec^8 x d(\sec x)\end{aligned}$$

Can we rewrite
 $\tan^4 x$ via $\sec x$?

Example

$$\begin{aligned}\int \tan^5 x \sec^9 x dx &= \int \tan^4 x \sec^8 x \tan x \sec x dx \\ &= \int \tan^4 x \sec^8 x d(\sec x) \\ &= \int (\tan^2 x)^2 \sec^8 x d(\sec x)\end{aligned}$$

Can we rewrite
 $\tan^4 x$ via $\sec x$?

Example

$$\begin{aligned}\int \tan^5 x \sec^9 x dx &= \int \tan^4 x \sec^8 x \tan x \sec x dx \\&= \int \tan^4 x \sec^8 x d(\sec x) \\&= \int (\tan^2 x)^2 \sec^8 x d(\sec x) \\&= \int (\sec^2 x - 1)^2 \sec^8 x d(\sec x)\end{aligned}$$

Can we rewrite
 $\tan^4 x$ via $\sec x$?

Example

$$\begin{aligned}
 \int \tan^5 x \sec^9 x dx &= \int \tan^4 x \sec^8 x \tan x \sec x dx \\
 &= \int \tan^4 x \sec^8 x d(\sec x) && \left| \begin{array}{l} \text{Can we rewrite} \\ \tan^4 x \text{ via } \sec x? \end{array} \right. \\
 &= \int (\tan^2 x)^2 \sec^8 x d(\sec x) \\
 &= \int (\sec^2 x - 1)^2 \sec^8 x d(\sec x) && \left| \text{Set } u = \sec x \right. \\
 &= \int (1 - u^2)^2 u^8 du
 \end{aligned}$$

Example

$$\begin{aligned}\int \tan^5 x \sec^9 x dx &= \int \tan^4 x \sec^8 x \tan x \sec x dx \\&= \int \tan^4 x \sec^8 x d(\sec x) && \left| \begin{array}{l} \text{Can we rewrite} \\ \tan^4 x \text{ via } \sec x? \end{array} \right. \\&= \int (\tan^2 x)^2 \sec^8 x d(\sec x) \\&= \int (\sec^2 x - 1)^2 \sec^8 x d(\sec x) && \left| \text{Set } u = \sec x \right. \\&= \int (1 - u^2)^2 u^8 du \\&= \int (1 - 2u^2 + u^4) u^8 du\end{aligned}$$

Example

$$\begin{aligned}\int \tan^5 x \sec^9 x dx &= \int \tan^4 x \sec^8 x \tan x \sec x dx \\&= \int \tan^4 x \sec^8 x d(\sec x) && \left| \begin{array}{l} \text{Can we rewrite} \\ \tan^4 x \text{ via } \sec x? \end{array} \right. \\&= \int (\tan^2 x)^2 \sec^8 x d(\sec x) \\&= \int (\sec^2 x - 1)^2 \sec^8 x d(\sec x) && \left| \begin{array}{l} \text{Set } u = \sec x \end{array} \right. \\&= \int (1 - u^2)^2 u^8 du \\&= \int (1 - 2u^2 + u^4) u^8 du \\&= \int (u^8 - 2u^{10} + u^{12}) du\end{aligned}$$

Example

$$\begin{aligned}\int \tan^5 x \sec^9 x dx &= \int \tan^4 x \sec^8 x \tan x \sec x dx \\&= \int \tan^4 x \sec^8 x d(\sec x) && \left| \begin{array}{l} \text{Can we rewrite} \\ \tan^4 x \text{ via } \sec x? \end{array} \right. \\&= \int (\tan^2 x)^2 \sec^8 x d(\sec x) \\&= \int (\sec^2 x - 1)^2 \sec^8 x d(\sec x) && \left| \begin{array}{l} \text{Set } u = \sec x \end{array} \right. \\&= \int (1 - u^2)^2 u^8 du \\&= \int (1 - 2u^2 + u^4) u^8 du \\&= \int (u^8 - 2u^{10} + u^{12}) du \\&= ?\end{aligned}$$

Example

$$\begin{aligned}\int \tan^5 x \sec^9 x dx &= \int \tan^4 x \sec^8 x \tan x \sec x dx \\&= \int \tan^4 x \sec^8 x d(\sec x) && \left| \begin{array}{l} \text{Can we rewrite} \\ \tan^4 x \text{ via } \sec x? \end{array} \right. \\&= \int (\tan^2 x)^2 \sec^8 x d(\sec x) \\&= \int (\sec^2 x - 1)^2 \sec^8 x d(\sec x) && \left| \text{Set } u = \sec x \right. \\&= \int (1 - u^2)^2 u^8 du \\&= \int (1 - 2u^2 + u^4) u^8 du \\&= \int (u^8 - 2u^{10} + u^{12}) du \\&= \frac{u^9}{9} - 2\frac{u^{11}}{11} + \frac{u^{13}}{13} + C\end{aligned}$$

Example

$$\begin{aligned}\int \tan^5 x \sec^9 x dx &= \int \tan^4 x \sec^8 x \tan x \sec x dx \\&= \int \tan^4 x \sec^8 x d(\sec x) && \left| \begin{array}{l} \text{Can we rewrite} \\ \tan^4 x \text{ via } \sec x? \end{array} \right. \\&= \int (\tan^2 x)^2 \sec^8 x d(\sec x) \\&= \int (\sec^2 x - 1)^2 \sec^8 x d(\sec x) && \left| \begin{array}{l} \text{Set } u = \sec x \end{array} \right. \\&= \int (1 - u^2)^2 u^8 du \\&= \int (1 - 2u^2 + u^4) u^8 du \\&= \int (u^8 - 2u^{10} + u^{12}) du \\&= \frac{u^9}{9} - 2\frac{u^{11}}{11} + \frac{u^{13}}{13} + C\end{aligned}$$

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 \end{aligned}$$

Example

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 \int \tan^5 x \sec^9 x dx &= \int \tan^4 x \sec^8 x \tan x \sec x dx \\
 &= \int \tan^4 x \sec^8 x d(\sec x) && \left| \begin{array}{l} \text{Can we rewrite} \\ \tan^4 x \text{ via } \sec x? \end{array} \right. \\
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 &= \int (u^8 - 2u^{10} + u^{12}) du \\
 &= \frac{u^9}{9} - 2\frac{u^{11}}{11} + \frac{u^{13}}{13} + C \\
 &= \frac{\sec^9 x}{9} - 2\frac{\sec^{11} x}{11} + \frac{\sec^{13} x}{13} + C .
 \end{aligned}$$

Partial strategy for fast evaluation of $\int \tan^m x \sec^n x dx$

$$\int \tan^m x \sec^n x dx \quad \left| \begin{array}{l} n - \text{even}, n \geq 2 \end{array} \right.$$

$$\int \tan^m x \sec^n x dx \quad \left| \begin{array}{l} m - \text{odd}, n \geq 1 \end{array} \right.$$

Partial strategy for fast evaluation of $\int \tan^m x \sec^n x dx$

$$\int \tan^m x \sec^n x dx = \int \tan^m x \sec^{n-2} x d(\tan x) \quad \left| \begin{array}{l} n - \text{even}, n \geq 2 \\ \sec^2 x dx \\ = d(\tan x) \end{array} \right.$$

$\int \tan^m x \sec^n x dx$	$m - \text{odd}, n \geq 1$
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Partial strategy for fast evaluation of $\int \tan^m x \sec^n x dx$

$$\begin{aligned}
 \int \tan^m x \sec^n x dx &= \int \tan^m x \sec^{n-2} x d(\tan x) \\
 &= \int \tan^m x (1 + \tan^2 x)^{\frac{n-2}{2}} d(\tan x)
 \end{aligned}
 \left| \begin{array}{l} n - \text{even}, n \geq 2 \\ \sec^2 x dx \\ = d(\tan x) \\ \text{Express } \sec x \\ \text{via } \tan x \end{array} \right.$$

$$\int \tan^m x \sec^n x dx \quad \left| \begin{array}{l} m - \text{odd}, n \geq 1 \end{array} \right.$$

Partial strategy for fast evaluation of $\int \tan^m x \sec^n x dx$

$$\begin{aligned} \int \tan^m x \sec^n x dx &= \int \tan^m x \sec^{n-2} x d(\tan x) \\ &= \int \tan^m x (1 + \tan^2 x)^{\frac{n-2}{2}} d(\tan x) \end{aligned}$$

$n - \text{even}, n \geq 2$

$\sec^2 x dx$

$= d(\tan x)$

Express $\sec x$

via $\tan x$

$$\int \tan^m x \sec^n x dx$$

$m - \text{odd}, n \geq 1$

Partial strategy for fast evaluation of $\int \tan^m x \sec^n x dx$

$\begin{aligned} \int \tan^m x \sec^n x dx &= \int \tan^m x \sec^{n-2} x d(\tan x) \\ &= \int \tan^m x (1 + \tan^2 x)^{\frac{n-2}{2}} d(\tan x) \\ &= \int u^m (1 + u^2)^{\frac{n-2}{2}} du \end{aligned}$	$n - \text{even}, n \geq 2$ $\sec^2 x dx = d(\tan x)$ Express $\sec x$ via $\tan x$ Set $u = \tan x$
$\int \tan^m x \sec^n x dx$	$m - \text{odd}, n \geq 1$

Partial strategy for fast evaluation of $\int \tan^m x \sec^n x dx$

$\begin{aligned}\int \tan^m x \sec^n x dx &= \int \tan^m x \sec^{n-2} x d(\tan x) \\ &= \int \tan^m x (1 + \tan^2 x)^{\frac{n-2}{2}} d(\tan x) \\ &= \int u^m (1 + u^2)^{\frac{n-2}{2}} du\end{aligned}$	$n - \text{even}, n \geq 2$ $\sec^2 x dx$ $= d(\tan x)$ Express $\sec x$ via $\tan x$ Set $u = \tan x$
$\int \tan^m x \sec^n x dx = \int \tan^{m-1} x \sec^{n-1} x d(\sec x)$	$m - \text{odd}, n \geq 1$ $\tan x \sec x dx$ $= d(\sec x)$

Partial strategy for fast evaluation of $\int \tan^m x \sec^n x dx$

$$\begin{aligned}
 \int \tan^m x \sec^n x dx &= \int \tan^m x \sec^{n-2} x d(\tan x) \\
 &= \int \tan^m x (1 + \tan^2 x)^{\frac{n-2}{2}} d(\tan x) \\
 &= \int u^m (1 + u^2)^{\frac{n-2}{2}} du
 \end{aligned}$$

$n - \text{even}, n \geq 2$

$\sec^2 x dx$

$= d(\tan x)$

Express $\sec x$
via $\tan x$

Set $u = \tan x$

$$\begin{aligned}
 \int \tan^m x \sec^n x dx &= \int \tan^{m-1} x \sec^{n-1} x d(\sec x) \\
 &= \int (\sec^2 x - 1)^{\frac{m-1}{2}} \sec^{n-1} x d(\sec x)
 \end{aligned}$$

$m - \text{odd}, n \geq 1$

$\tan x \sec x dx$

$= d(\sec x)$

Express $\tan x$
via $\sec x$

Partial strategy for fast evaluation of $\int \tan^m x \sec^n x dx$

$\begin{aligned}\int \tan^m x \sec^n x dx &= \int \tan^m x \sec^{n-2} x d(\tan x) \\ &= \int \tan^m x (1 + \tan^2 x)^{\frac{n-2}{2}} d(\tan x) \\ &= \int u^m (1 + u^2)^{\frac{n-2}{2}} du\end{aligned}$	$n - \text{even}, n \geq 2$ $\sec^2 x dx$ $= d(\tan x)$ Express $\sec x$ via $\tan x$ Set $u = \tan x$
$\begin{aligned}\int \tan^m x \sec^n x dx &= \int \tan^{m-1} x \sec^{n-1} x d(\sec x) \\ &= \int (\sec^2 x - 1)^{\frac{m-1}{2}} \sec^{n-1} x d(\sec x)\end{aligned}$	$m - \text{odd}, n \geq 1$ $\tan x \sec x dx$ $= d(\sec x)$ Express $\tan x$ via $\sec x$

Partial strategy for fast evaluation of $\int \tan^m x \sec^n x dx$

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$\begin{aligned} \int \tan^m x \sec^n x dx &= \int \tan^{m-1} x \sec^{n-1} x d(\sec x) \\ &= \int (\sec^2 x - 1)^{\frac{m-1}{2}} \sec^{n-1} x d(\sec x) \end{aligned}$	$m - \text{odd}, n \geq 1$ $\tan x \sec x dx$ $= d(\sec x)$ Express $\tan x$ via $\sec x$

Partial strategy for fast evaluation of $\int \tan^m x \sec^n x dx$

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$\begin{aligned} \int \tan^m x \sec^n x dx &= \int \tan^{m-1} x \sec^{n-1} x d(\sec x) \\ &= \int (\sec^2 x - 1)^{\frac{m-1}{2}} \sec^{n-1} x d(\sec x) \\ &= \int (u^2 - 1)^{\frac{m-1}{2}} u^n du \end{aligned}$	$m - \text{odd}, n \geq 1$ $\tan x \sec x dx$ $= d(\sec x)$ Express $\tan x$ via $\sec x$ Set $u = \sec x$

Partial strategy for fast evaluation of $\int \tan^m x \sec^n x dx$

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$\begin{aligned}\int \tan^m x \sec^n x dx &= \int \tan^{m-1} x \sec^{n-1} x d(\sec x) \\ &= \int (\sec^2 x - 1)^{\frac{m-1}{2}} \sec^{n-1} x d(\sec x) \\ &= \int (u^2 - 1)^{\frac{m-1}{2}} u^n du\end{aligned}$	$m - \text{odd}, n \geq 1$ $\tan x \sec x dx = d(\sec x)$ Express $\tan x$ via $\sec x$ Set $u = \sec x$
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Outside of the above cases we either use more tricks or resort to the general method $x = 2 \arctan t$.

Partial strategy for fast evaluation of $\int \tan^m x \sec^n x dx$

$\begin{aligned}\int \tan^m x \sec^n x dx &= \int \tan^m x \sec^{n-2} x d(\tan x) \\ &= \int \tan^m x (1 + \tan^2 x)^{\frac{n-2}{2}} d(\tan x) \\ &= \int u^m (1 + u^2)^{\frac{n-2}{2}} du\end{aligned}$	$n - \text{even}, n \geq 2$ $\sec^2 x dx$ $= d(\tan x)$ Express $\sec x$ via $\tan x$ Set $u = \tan x$
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Outside of the above cases we either use **more tricks** or resort to **the general method** $x = 2 \arctan t$.

Example

$$\int \tan x dx$$

Example

Example

$$\int \tan x dx = \int \frac{\sin x}{\cos x} dx$$

Example

Example

$$\int \tan x dx = \int \frac{\sin x}{\cos x} dx = \int \frac{1}{\cos x} d(?)$$

Example

Example

$$\int \tan x dx = \int \frac{\sin x}{\cos x} dx = \int \frac{1}{\cos x} d(-\cos x)$$

Example

Example

$$\begin{aligned}\int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{1}{\cos x} d(-\cos x) \quad \Bigg| \quad \text{Set } u = \cos x \\ &= - \int \frac{du}{u}\end{aligned}$$

Example

Example

$$\begin{aligned}\int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{1}{\cos x} d(-\cos x) \quad \Bigg| \quad \text{Set } u = \cos x \\ &= - \int \frac{du}{u}\end{aligned}$$

Example

Example

$$\begin{aligned}\int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{1}{\cos x} d(-\cos x) \quad \left| \text{Set } u = \cos x \right. \\ &= - \int \frac{du}{u} = -\ln |u| + C\end{aligned}$$

Example

Example

$$\begin{aligned}\int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{1}{\cos x} d(-\cos x) && \left| \text{Set } u = \cos x \right. \\ &= - \int \frac{du}{u} = -\ln |u| + C \\ &= -\ln |\cos x| + C\end{aligned}$$

Example

Example

$$\begin{aligned}\int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{1}{\cos x} d(-\cos x) && \left| \text{Set } u = \cos x \right. \\ &= - \int \frac{du}{u} = -\ln |u| + C \\ &= -\ln |\text{red } x| + C = \ln |\text{red } x| + C\end{aligned}$$

Example

Example

$$\begin{aligned}\int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{1}{\cos x} d(-\cos x) \quad \Bigg| \quad \text{Set } u = \cos x \\ &= - \int \frac{du}{u} = -\ln |u| + C \\ &= -\ln |\cos x| + C = \ln |\sec x| + C\end{aligned}$$

The following can be/was computed via $x = 2 \arctan t$.

Example

$$\int \sec x dx$$

Example

$$\begin{aligned}\int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{1}{\cos x} d(-\cos x) \quad \Bigg| \quad \text{Set } u = \cos x \\ &= - \int \frac{du}{u} = -\ln |u| + C \\ &= -\ln |\cos x| + C = \ln |\sec x| + C\end{aligned}$$

The following can be/was computed via $x = 2 \arctan t$. Alternatively:

Example

$$\int \sec x dx = \int \sec x \frac{(\sec x + \tan x)}{(\sec x + \tan x)} dx$$

Example

$$\begin{aligned}\int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{1}{\cos x} d(-\cos x) \quad \Bigg| \quad \text{Set } u = \cos x \\ &= - \int \frac{du}{u} = -\ln |u| + C \\ &= -\ln |\cos x| + C = \ln |\sec x| + C\end{aligned}$$

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$$\begin{aligned}\int \sec x dx &= \int \sec x \frac{(\sec x + \tan x)}{(\sec x + \tan x)} dx \\ &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx\end{aligned}$$

Example

$$\begin{aligned}
 \int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{1}{\cos x} d(-\cos x) \quad \Bigg| \quad \text{Set } u = \cos x \\
 &= - \int \frac{du}{u} = -\ln |u| + C \\
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 &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx \\
 &= \int \frac{d(\tan x + \sec x)}{\sec x + \tan x}
 \end{aligned}$$

Example

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 \int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{1}{\cos x} d(-\cos x) \quad \Bigg| \quad \text{Set } u = \cos x \\
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 &= \int \frac{du}{u}
 \end{aligned}$$

Example

$$\begin{aligned}
 \int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{1}{\cos x} d(-\cos x) && \left| \text{Set } u = \cos x \right. \\
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 &= \int \frac{du}{u} = \ln |u| + C
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 \int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{1}{\cos x} d(-\cos x) && \left| \text{Set } u = \cos x \right. \\
 &= - \int \frac{du}{u} = -\ln |u| + C \\
 &= -\ln |\cos x| + C = \ln |\sec x| + C
 \end{aligned}$$

The following can be/was computed via $x = 2 \arctan t$. Alternatively:

Example

$$\begin{aligned}
 \int \sec x dx &= \int \sec x \frac{(\sec x + \tan x)}{(\sec x + \tan x)} dx \\
 &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx \\
 &= \int \frac{d(\tan x + \sec x)}{\sec x + \tan x} && \left| \text{Set } u = \sec x + \tan x \right. \\
 &= \int \frac{du}{u} = \ln |u| + C \\
 &= \ln |\sec x + \tan x| + C.
 \end{aligned}$$

Example

$$\int \tan^3 x dx$$

Example

$$\int \tan^3 x dx = \int \tan x \tan^2 x dx$$

Example

$$\begin{aligned}\int \tan^3 x dx &= \int \tan x \tan^2 x dx \\ &= \int \tan x (\sec^2 x - 1) dx\end{aligned}$$

Example

$$\begin{aligned}\int \tan^3 x dx &= \int \tan x \tan^2 x dx \\ &= \int \tan x (\sec^2 x - 1) dx \\ &= \int \tan x \sec^2 x dx - \int \tan x dx\end{aligned}$$

Example

$$\begin{aligned}\int \tan^3 x dx &= \int \tan x \tan^2 x dx \\ &= \int \tan x (\sec^2 x - 1) dx \\ &= \int \tan x \sec^2 x dx - \int \tan x dx\end{aligned}$$

Example

$$\begin{aligned}\int \tan^3 x dx &= \int \tan x \tan^2 x dx \\&= \int \tan x (\sec^2 x - 1) dx \\&= \int \tan x \sec^2 x dx - \int \tan x dx \\&= \int \tan x d(?) - ?\end{aligned}$$

Example

$$\begin{aligned}\int \tan^3 x dx &= \int \tan x \tan^2 x dx \\&= \int \tan x (\sec^2 x - 1) dx \\&= \int \tan x \sec^2 x dx - \int \tan x dx \\&= \int \tan x d(\tan x) - ?\end{aligned}$$

Example

$$\begin{aligned}\int \tan^3 x dx &= \int \tan x \tan^2 x dx \\&= \int \tan x (\sec^2 x - 1) dx \\&= \int \tan x \sec^2 x dx - \int \tan x dx \\&= \int \tan x d(\tan x) - ?\end{aligned}$$

Example

$$\begin{aligned}\int \tan^3 x dx &= \int \tan x \tan^2 x dx \\&= \int \tan x (\sec^2 x - 1) dx \\&= \int \tan x \sec^2 x dx - \int \tan x dx \\&= \int \tan x d(\tan x) - \ln |\sec x|\end{aligned}$$

Example

$$\begin{aligned}\int \tan^3 x dx &= \int \tan x \tan^2 x dx \\&= \int \tan x (\sec^2 x - 1) dx \\&= \int \tan x \sec^2 x dx - \int \tan x dx \\&= \int \tan x d(\tan x) - \ln |\sec x| \quad \left| \text{Set } u = \tan x \right. \\&= \int u du + \ln \left| \frac{1}{\sec x} \right|\end{aligned}$$

Example

$$\begin{aligned}\int \tan^3 x dx &= \int \tan x \tan^2 x dx \\&= \int \tan x (\sec^2 x - 1) dx \\&= \int \tan x \sec^2 x dx - \int \tan x dx \\&= \int \tan x d(\tan x) - \ln |\sec x| \quad \left| \text{Set } u = \tan x \right. \\&= \int u du + \ln \left| \frac{1}{\sec x} \right|\end{aligned}$$

Example

$$\begin{aligned}\int \tan^3 x dx &= \int \tan x \tan^2 x dx \\&= \int \tan x (\sec^2 x - 1) dx \\&= \int \tan x \sec^2 x dx - \int \tan x dx \\&= \int \tan x d(\tan x) - \ln |\sec x| \quad \left| \text{Set } u = \tan x \right. \\&= \int u du + \ln \left| \frac{1}{\sec x} \right| \\&= \frac{u^2}{2} + \ln |\cos x| + C\end{aligned}$$

Example

$$\begin{aligned}\int \tan^3 x dx &= \int \tan x \tan^2 x dx \\&= \int \tan x (\sec^2 x - 1) dx \\&= \int \tan x \sec^2 x dx - \int \tan x dx \\&= \int \tan x d(\tan x) - \ln |\sec x| \quad \left| \text{Set } u = \tan x \right. \\&= \int u du + \ln \left| \frac{1}{\sec x} \right| \\&= \frac{u^2}{2} + \ln |\cos x| + C\end{aligned}$$

Example

$$\begin{aligned}\int \tan^3 x dx &= \int \tan x \tan^2 x dx \\&= \int \tan x (\sec^2 x - 1) dx \\&= \int \tan x \sec^2 x dx - \int \tan x dx \\&= \int \tan x d(\tan x) - \ln |\sec x| \quad \left| \text{Set } u = \tan x \right. \\&= \int u du + \ln \left| \frac{1}{\sec x} \right| \\&= \frac{u^2}{2} + \ln |\cos x| + C \\&= \frac{\tan^2 x}{2} + \ln |\cos x| + C\end{aligned}$$

Example

$$\int \sec^3 x dx$$

Example

$$\int \sec^3 x dx = \int \sec x \sec^2 x dx$$

Example

$$\begin{aligned}\int \sec^3 x dx &= \int \sec x \sec^2 x dx \\ &= \int \sec x d(?)\end{aligned}$$

Example

$$\begin{aligned}\int \sec^3 x dx &= \int \sec x \sec^2 x dx \\ &= \int \sec x d(\tan x)\end{aligned}$$

Example

$$\int \sec^3 x dx = \int \sec x \sec^2 x dx$$

$$= \int \sec x d(\tan x)$$

$$= \sec x \tan x - \int \tan x d(\sec x)$$

Integrate
by parts

Example

$$\int \sec^3 x dx = \int \sec x \sec^2 x dx$$

$$= \int \sec x d(\tan x)$$

$$= \sec x \tan x - \int \tan x d(\sec x)$$

Integrate
by parts

Example

$$\int \sec^3 x dx = \int \sec x \sec^2 x dx$$

$$= \int \sec x d(\tan x)$$

$$= \sec x \tan x - \int \tan x d(\sec x)$$

$$= \sec x \tan x - \int \tan x \, dx$$

Integrate
by parts

Example

$$\begin{aligned}\int \sec^3 x dx &= \int \sec x \sec^2 x dx \\&= \int \sec x d(\tan x) \\&= \sec x \tan x - \int \tan x d(\sec x) \\&= \sec x \tan x - \int \tan x \sec x \tan x dx\end{aligned}$$

Integrate
by parts

Example

$$\begin{aligned}\int \sec^3 x dx &= \int \sec x \sec^2 x dx \\&= \int \sec x d(\tan x) \\&= \sec x \tan x - \int \tan x d(\sec x) \\&= \sec x \tan x - \int \tan x \sec x \tan x dx\end{aligned}$$

Integrate
by parts

Example

$$\begin{aligned}\int \sec^3 x dx &= \int \sec x \sec^2 x dx \\ &= \int \sec x d(\tan x) \\ &= \sec x \tan x - \int \tan x d(\sec x) \\ &= \sec x \tan x - \int \tan^2 x \sec x dx\end{aligned}$$

Integrate
by parts

Example

$$\begin{aligned}\int \sec^3 x dx &= \int \sec x \sec^2 x dx \\&= \int \sec x d(\tan x) \\&= \sec x \tan x - \int \tan x d(\sec x) \\&= \sec x \tan x - \int \tan^2 x \sec x dx \\&= \sec x \tan x - \int (\sec^2 x - 1) \sec x dx\end{aligned}$$

Integrate
by parts

Example

$$\begin{aligned}\int \sec^3 x dx &= \int \sec x \sec^2 x dx \\&= \int \sec x d(\tan x) \\&= \sec x \tan x - \int \tan x d(\sec x) \\&= \sec x \tan x - \int \tan^2 x \sec x dx \\&= \sec x \tan x - \int (\sec^2 x - 1) \sec x dx \\&= \sec x \tan x - \int \sec^3 x dx + \int \sec x dx\end{aligned}$$

Integrate
by parts

Example

$$\begin{aligned}\int \sec^3 x dx &= \int \sec x \sec^2 x dx \\&= \int \sec x d(\tan x) \\&= \sec x \tan x - \int \tan x d(\sec x) \\&= \sec x \tan x - \int \tan^2 x \sec x dx \\&= \sec x \tan x - \int (\sec^2 x - 1) \sec x dx \\&= \sec x \tan x - \int \sec^3 x dx + \int \sec x dx\end{aligned}$$

Integrate
by parts

Example

$$\begin{aligned}\int \sec^3 x dx &= \int \sec x \sec^2 x dx \\&= \int \sec x d(\tan x) \\&= \sec x \tan x - \int \tan x d(\sec x) \\&= \sec x \tan x - \int \tan^2 x \sec x dx \\&= \sec x \tan x - \int (\sec^2 x - 1) \sec x dx \\&= \sec x \tan x - \int \sec^3 x dx + \int \sec x dx\end{aligned}$$

Integrate
by parts

Example

$$\begin{aligned}
 \int \sec^3 x dx &= \int \sec x \sec^2 x dx \\
 &= \int \sec x d(\tan x) \\
 &= \sec x \tan x - \int \tan x d(\sec x) \\
 &= \sec x \tan x - \int \tan^2 x \sec x dx \\
 &= \sec x \tan x - \int (\sec^2 x - 1) \sec x dx \\
 &= \sec x \tan x - \int \sec^3 x dx + \int \sec x dx \\
 2 \int \sec^3 x dx &= \sec x \tan x + ? \quad + C
 \end{aligned}$$

Integrate
by parts

Example

$$\begin{aligned}\int \sec^3 x dx &= \int \sec x \sec^2 x dx \\&= \int \sec x d(\tan x) && \left| \begin{array}{l} \text{Integrate} \\ \text{by parts} \end{array} \right. \\&= \sec x \tan x - \int \tan x d(\sec x) \\&= \sec x \tan x - \int \tan^2 x \sec x dx \\&= \sec x \tan x - \int (\sec^2 x - 1) \sec x dx \\&= \sec x \tan x - \int \sec^3 x dx + \int \sec x dx \\2 \int \sec^3 x dx &= \sec x \tan x + \textcolor{red}{?} + C\end{aligned}$$

Example

$$\begin{aligned}
 \int \sec^3 x dx &= \int \sec x \sec^2 x dx \\
 &= \int \sec x d(\tan x) \\
 &= \sec x \tan x - \int \tan x d(\sec x) \\
 &= \sec x \tan x - \int \tan^2 x \sec x dx \\
 &= \sec x \tan x - \int (\sec^2 x - 1) \sec x dx \\
 &= \sec x \tan x - \int \sec^3 x dx + \int \sec x dx \\
 2 \int \sec^3 x dx &= \sec x \tan x + \ln |\sec x + \tan x| + C
 \end{aligned}$$

Integrate
by parts

Example

$$\int \sec^3 x dx = \int \sec x \sec^2 x dx$$

$$= \int \sec x d(\tan x)$$

Integrate
by parts

$$= \sec x \tan x - \int \tan x d(\sec x)$$

$$= \sec x \tan x - \int \tan^2 x \sec x dx$$

$$= \sec x \tan x - \int (\sec^2 x - 1) \sec x dx$$

$$= \sec x \tan x - \int \sec^3 x dx + \int \sec x dx$$

$$2 \int \sec^3 x dx = \sec x \tan x + \ln |\sec x + \tan x| + C$$

$$\int \sec^3 x dx = \frac{1}{2} (\sec x \tan x + \ln |\sec x + \tan x|) + K.$$

To evaluate integrals of the form

$$\textcircled{1} \int \sin(mx) \cos(nx) dx$$

$$\textcircled{2} \int \sin(mx) \sin(nx) dx$$

$$\textcircled{3} \int \cos(mx) \cos(nx) dx$$

use the corresponding identity:

$$\textcircled{1} \sin A \cos B = \frac{1}{2} [\sin(A - B) + \sin(A + B)]$$

$$\textcircled{2} \sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\textcircled{3} \cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

Example

$$\int \sin(4x) \cos(5x) dx$$

Example

$$\int \sin(4x) \cos(5x) dx = \int \frac{1}{2} [\sin(4x - 5x) + \sin(4x + 5x)] dx$$

Example

$$\begin{aligned}\int \sin(4x) \cos(5x) dx &= \int \frac{1}{2} [\sin(4x - 5x) + \sin(4x + 5x)] dx \\ &= \frac{1}{2} \int (\sin(-x) + \sin(9x)) dx\end{aligned}$$

Example

$$\begin{aligned}\int \sin(4x) \cos(5x) dx &= \int \frac{1}{2} [\sin(4x - 5x) + \sin(4x + 5x)] dx \\ &= \frac{1}{2} \int (\sin(-x) + \sin(9x)) dx \\ &= \frac{1}{2} \int (-\sin x + \sin(9x)) dx\end{aligned}$$

Example

$$\begin{aligned}\int \sin(4x) \cos(5x) dx &= \int \frac{1}{2} [\sin(4x - 5x) + \sin(4x + 5x)] dx \\ &= \frac{1}{2} \int (\sin(-x) + \sin(9x)) dx \\ &= \frac{1}{2} \int (-\sin x + \sin(9x)) dx \\ &= \frac{1}{2} \left(\cos x - \frac{1}{9} \cos(9x) \right) + C\end{aligned}$$